MuSR Blackboard Population Study 01

First state-resolved analysis of control efficiency

Control Efficiency of MAS Systems

September 2026

Abstract

This report analyzes the first population-scale MuSR Team Allocation experiment using the frozen night–dawn–day blackboard-control protocol. The main analysis is the $q = 1$ study with 270 episodes: one task (`task_001`), three persistence values $\rho \in \{0.74, 0.85, 1.00\}$, four control budgets $b \in \{3, 6, 12, 24\}$, truth and false control, and no-control baselines. The emphasis is on the same control observables used in the earlier program: state-local susceptibility χ, controller-to-population transfer information $T_\pi = I(U_k; n_{k+1} \mid n_k)$, information fraction η_{IF}, information-response efficiency η_{IR}, and controlled current J_c. The results show strong persistence dependence, non-monotone budget dependence, and sharply state-localized control response. The false controller disrupts but usually does not capture the population; its strongest regime occurs at low persistence. The current package does not identify the thermodynamic efficiency η_{th} because the single-affinity h calibration is unsupported for the new dawn-blackboard actuator.

1 Main conclusions

- The main $q = 1$ experiment completed all 270 planned episodes (27 cells × 10 repetitions, 30 rounds each). The supplied analysis archive additionally contains 120 exploratory false-control episodes with $q = 3$, for 390 episodes total.
- The autonomous dynamics are strongly truth-seeking: no-control final truth is 92.5%, 99.2%, and 100.0% at $\rho = 0.74, 0.85, 1.00$ respectively.
- False control is therefore mostly disruptive rather than population-capturing. Across all standard $q = 1$ false-control cells, the false target wins 8/120 episodes. However, the effect is strongly persistence-dependent: at $\rho = 0.74, b = 12$, final truth falls to 57.9%, the false target reaches 36.2%, and the false target dominates in 3/10 episodes.
- The persistence crossover appears shifted to lower ρ relative to the earlier peer-only persistence dynamics. This is consistent with the blackboard adding an information-recovery pathway: inactive evidence can be reintroduced through later REPORTs. With only three ρ values, this should be described as a shifted crossover rather than a measured critical point.
- The control signal at the next macro-step is clearly measurable. Whole-cell T_π reaches 0.316 bits for truth control at $(\rho, b) = (0.74, 12)$ and 0.310 bits for false control at $(0.85, 24)$. The state-resolved maps contain much larger local values.
- Control is not monotone in budget. Both outcomes and information/response quantities show intermediate-budget structure. In particular, at $\rho = 0.74$ false control is stronger at $b = 12$ than at $b = 24$, while excessive truth-directed control can degrade the otherwise successful autonomous dynamics.

2 Study configuration

Table 1: Main parameters of the frozen $q = 1$ population study.

Parameter	Value
Task	MuSR Team Allocation, `task_001`; gold semantic target `ALLOCATION_0`; false target `ALLOCATION_1`
Model	`gwdg/openai-gpt-oss-120b`, temperature 1.0
Population	$N = 24$ ordinary agents; coordinator excluded from population counts and sensing pool
Horizon	30 macro-rounds per episode
Repetitions	10 per structural cell
Social channel	Blackboard, uniform sampling, self-authored messages excluded
Social bandwidth	$q = 1$ live board message sampled per focal update
Board lifetime	$\tau_B = 1$ round
Persistence	$\rho \in \{0.74, 0.85, 1.00\}$ acting on active evidence memory
Controller sensing	$q_c = 12$ ordinary-agent votes sampled at night
Controller policy	soft target, $\beta = 4$, $\theta = 0.5$
Controller timing	`night_dawn_autonomous_day_v1`; control acts at dawn only
Budgets	$b \in \{3, 6, 12, 24\}$ DIRECTIVEs seeded at dawn when $U_k = 1$
Control arms	no control; truth control; false control
Initial evidence	Frozen F9 $N = 24$ assignment; one canonical F9 evidence card / one latent value per ordinary agent
Main $q = 1$ size	3 no-control cells + 12 truth-control + 12 false-control = 27 cells = 270 episodes
Exploratory extension	False control with $q = 3$: 12 cells, 120 episodes

The causal timing retained for the core information measure is

$$n_k \longrightarrow Y_k \longrightarrow P(U_k \mid Y_k) \longrightarrow U_k \longrightarrow B_k^{\text{dawn}} \longrightarrow \text{autonomous day} \longrightarrow n_{k+1}.$$

For the state-resolved plots, x denotes the current *controller-target fraction*: it is the truth fraction in truth-control cells and the fixed false-target fraction in false-control cells.

3 Behavioral phase structure

Figure 1 shows the coarse (ρ, b) outcome landscape. Persistence is the dominant robustness coordinate. At $\rho = 1$, truth essentially saturates even under false control; at $\rho = 0.85$, appreciable false steering appears only at the largest budget; at $\rho = 0.74$, multiple budgets materially perturb the population, with the strongest false-target outcome at $b = 12$ rather than $b = 24$.

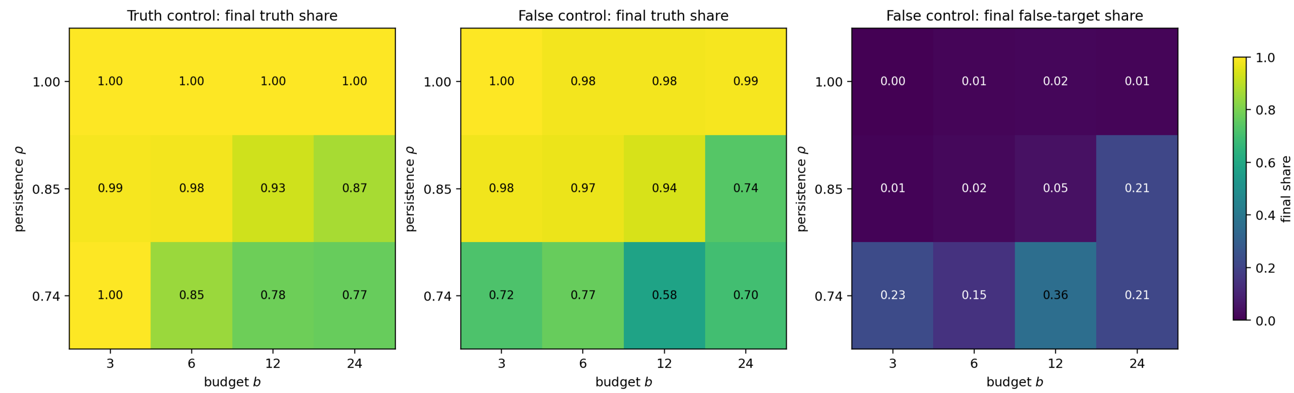

Figure 1: Behavioral (ρ, b) phase diagrams for the main $q = 1$ study. Values are final-round means over 10 episodes per cell.

Table 2: Truth-control outcomes. The no-control column is the matching persistence baseline and is repeated across budgets only for comparison.

ρ	b	No-control final truth	Truth-control final truth
0.74	3	92.5%	100.0%
0.74	6	92.5%	85.0%
0.74	12	92.5%	77.9%
0.74	24	92.5%	76.7%
0.85	3	99.2%	98.8%
0.85	6	99.2%	98.3%
0.85	12	99.2%	92.9%
0.85	24	99.2%	87.1%
1.00	3	100.0%	100.0%
1.00	6	100.0%	100.0%
1.00	12	100.0%	100.0%
1.00	24	100.0%	100.0%

Table 3: False-control outcomes. "False wins" means the fixed false semantic target is the unique dominant final option.

ρ	b	No-control final truth	False-control final truth	Final false target	False wins
0.74	3	92.5%	72.1%	22.9%	2/10
0.74	6	92.5%	76.7%	14.6%	1/10
0.74	12	92.5%	57.9%	36.2%	3/10
0.74	24	92.5%	69.6%	21.2%	1/10
0.85	3	99.2%	97.9%	0.8%	0/10
0.85	6	99.2%	97.1%	2.1%	0/10
0.85	12	99.2%	93.8%	4.6%	0/10
0.85	24	99.2%	73.8%	21.2%	1/10
1.00	3	100.0%	100.0%	0.0%	0/10
1.00	6	100.0%	98.3%	1.2%	0/10
1.00	12	100.0%	98.3%	1.7%	0/10
1.00	24	100.0%	98.8%	1.2%	0/10

The aggregate statement "false control mostly fails to capture the population" and the persistence-resolved statement “a low-persistence controllable regime exists” are both true. Aggregating across ρ hides the latter. This is why the remainder of the report keeps the three persistence slices explicit.

4 Whole-cell control phase diagrams

The core whole-cell metrics are shown in Figure 2. We retain the established estimators from the previous studies:

$$T_\pi = I(U_k; n_{k+1} \mid n_k),$$

with χ the state-matched target-fraction response, η_{IF} the repository's existing target information-fraction estimator, and η_{IR} the corrected occupancy-weighted information-response efficiency. The controlled current J_c is also retained.

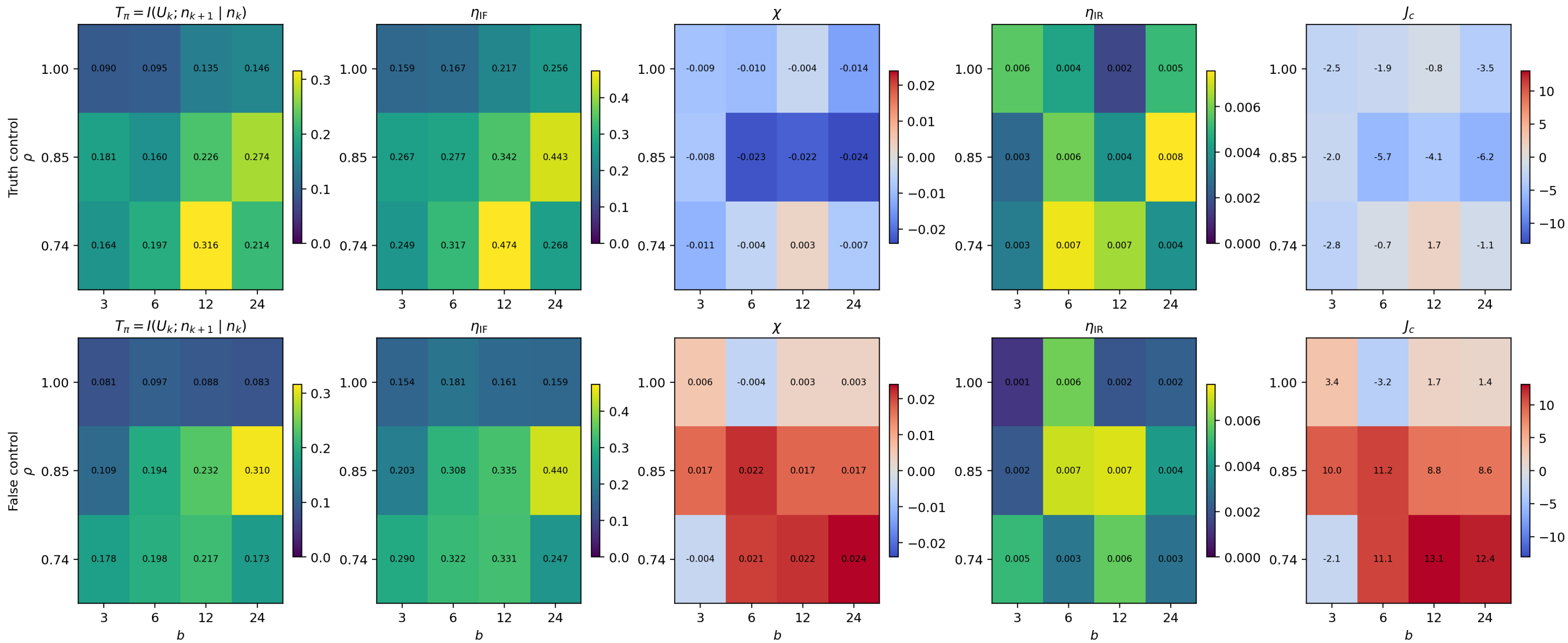

Figure 2: Whole-cell (ρ, b) landscapes for truth control (top row) and false control (bottom row), $q = 1$. The columns are T_π, $\eta_{\rm IF}$, χ, $\eta_{\rm IR}$, and J_c.

Several features are already visible. Truth control at $\rho = 0.74$ has a pronounced information-transfer maximum at $b = 12$ ($T_\pi = 0.316$ bits, $\eta_{\mathrm{IF}} = 0.474$), whereas false control at $\rho = 0.85$ reaches its largest whole-cell transfer at $b = 24$ ($T_\pi = 0.310$ bits, $\eta_{\mathrm{IF}} = 0.440$). The signed response and current need not peak where the transfer information peaks: information that the controller acted is not synonymous with motion in the desired direction.

5 State-resolved $x \times b$ phase diagrams

The state-resolved maps are the main diagnostic of control efficiency. The supplied aggregate phase-map CSV had a structural-join issue, so the maps below were reconstructed directly from the valid state-resolved rows in `primary_estimates.csv` and `efficiencies.csv`. Unsupported cells are left blank. The rightmost panel in each row is a descriptive ρ-aggregate weighted by the number of state observations. It is not a substitute for the separate persistence slices.

5.1 T_π: controller-to-population transfer information

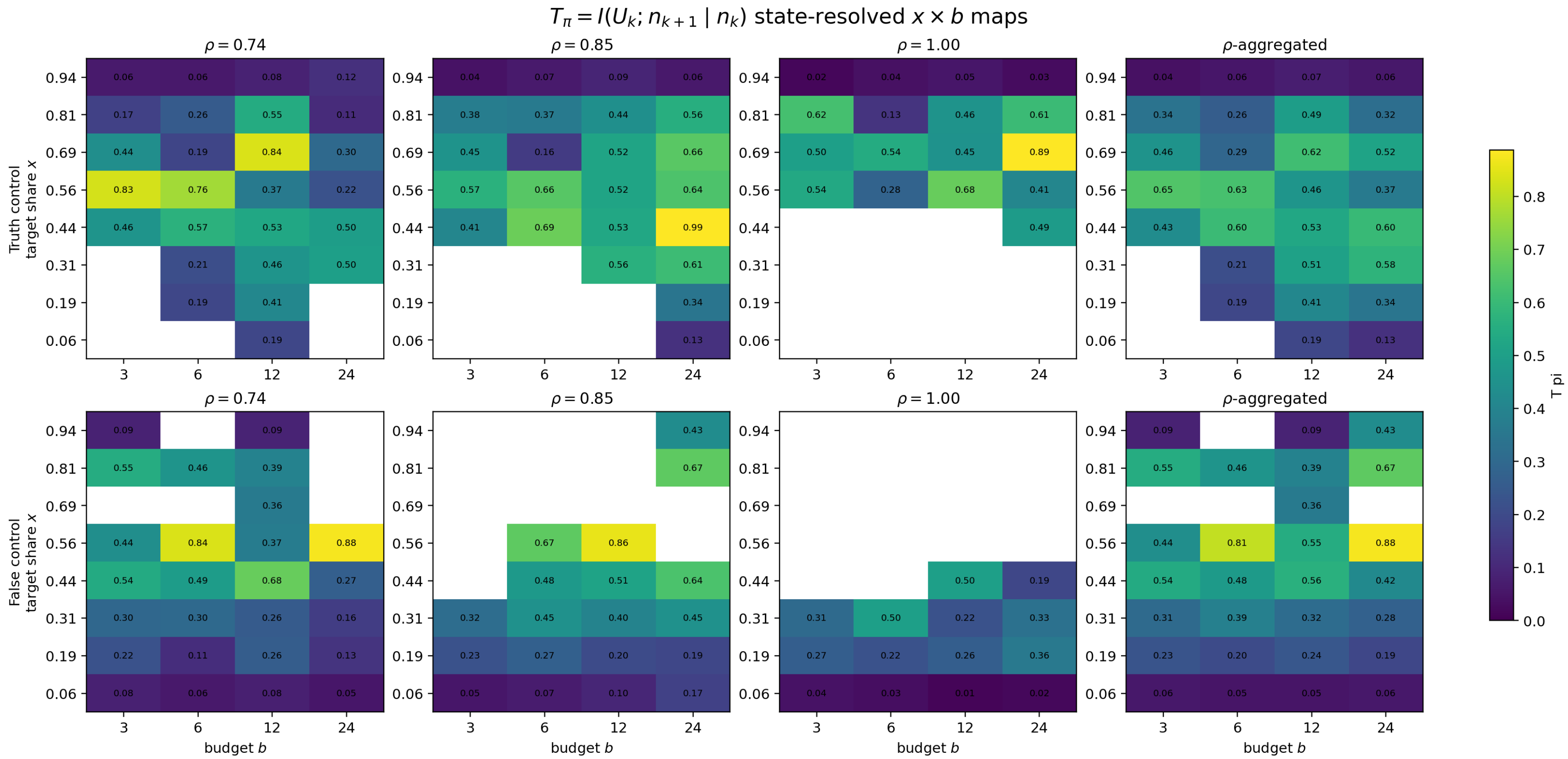

Figure 3: State-resolved $T_\pi(x, b)$ in bits. Top: truth-control target share x is truth share. Bottom: false-control target share x is the fixed false-target share. Each row shows $\rho = 0.74$, 0.85, 1.00, and an observation-weighted aggregate.

The state-local signal is much stronger than the whole-cell average. In truth control, intermediate x often supports large T_π, while near-saturated truth states can have much smaller transfer. In false control, the strongest information-transfer regions shift with persistence and budget. This is why a single cell average can obscure the control landscape.

5.2 η_{IF}: information fraction

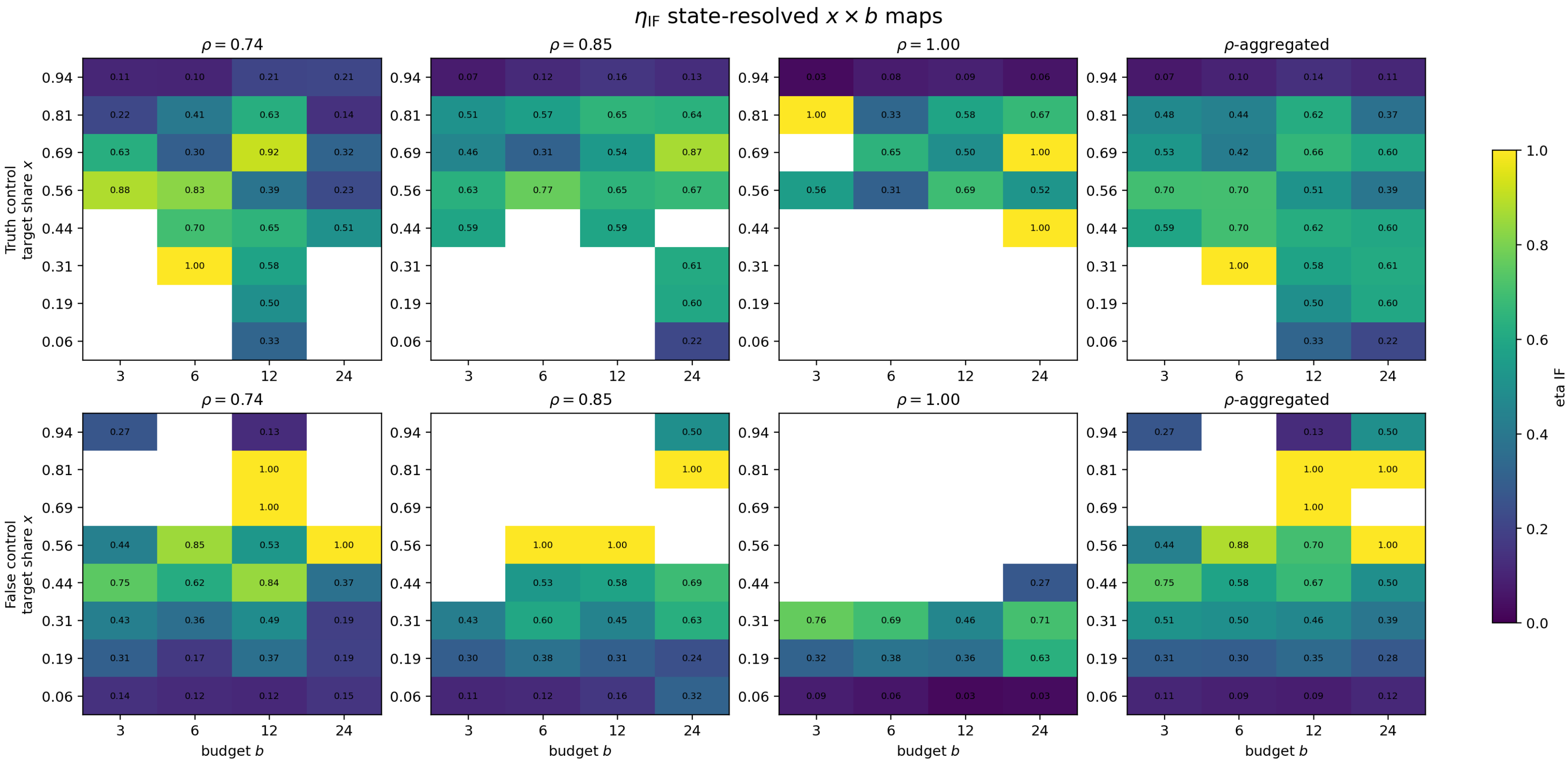

Figure 4: State-resolved information fraction $\eta_{\mathrm{IF}}(x, b)$ for the three persistence slices and the descriptive ρ-aggregate.

5.3 χ: signed susceptibility

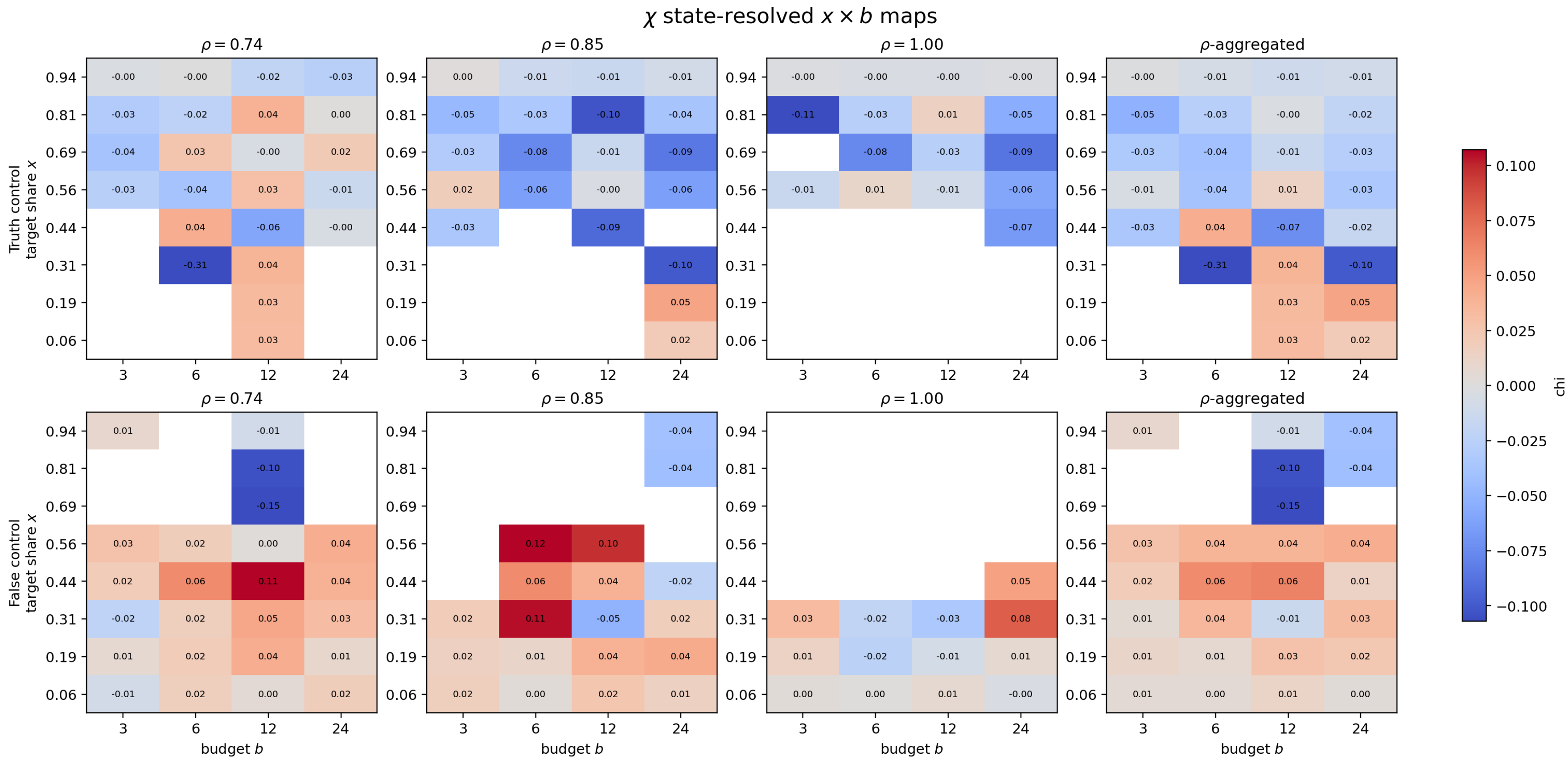

Figure 5: State-resolved signed susceptibility $\chi(x, b)$. Positive values move the population toward the configured controller target; negative values move away from it. Blank cells lack supported state/action coverage.

The susceptibility maps demonstrate why the budget response is non-monotone. The same b can have opposite response signs at different population states. For false control at lower persistence, there are positive-susceptibility regions around intermediate false-target shares, whereas truth control contains several negative-response pockets despite the global truth-seeking dynamics.

5.4 η_{IR}: information-response efficiency

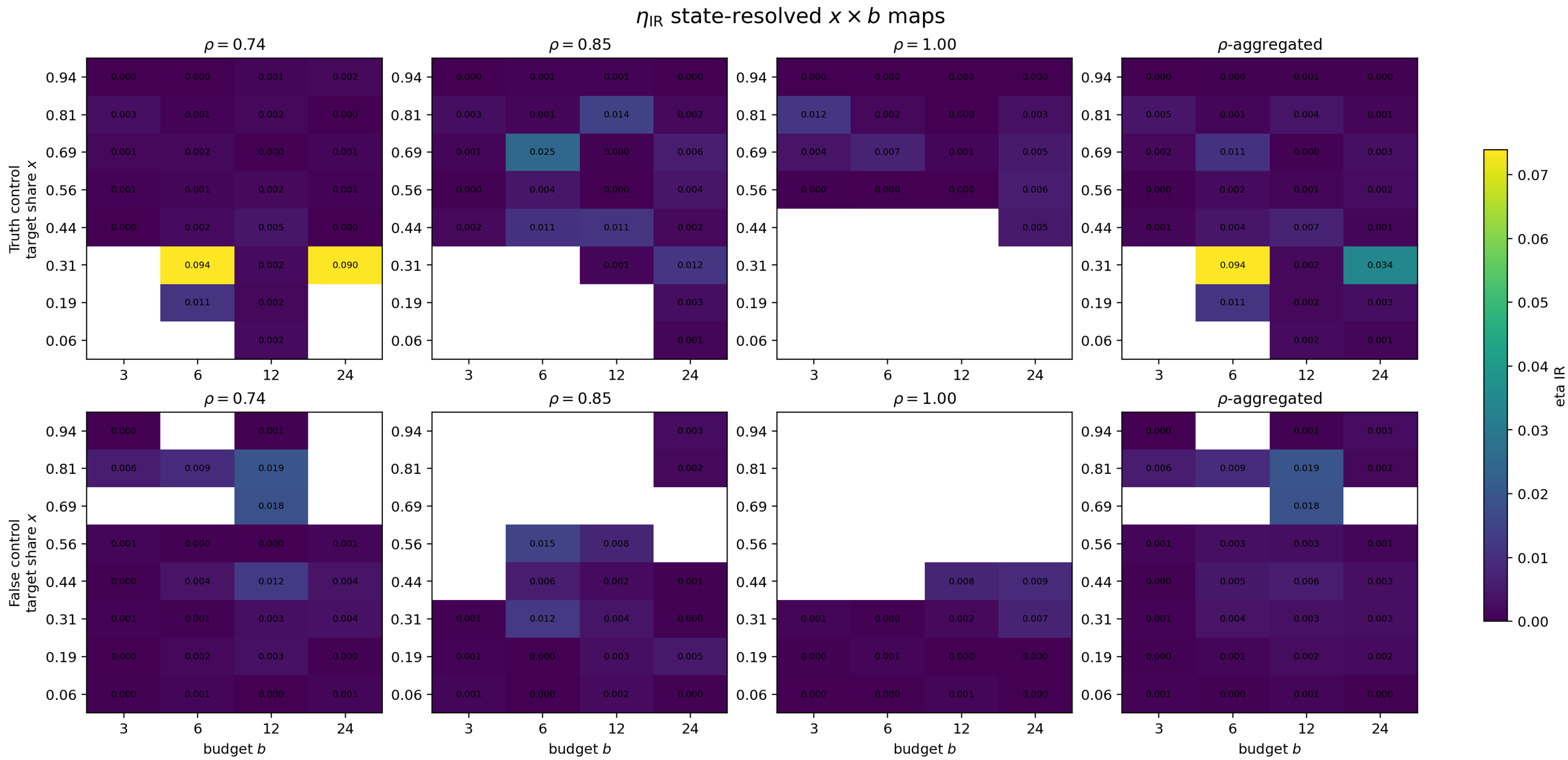

Figure 6: State-resolved $\eta_{\mathrm{IR}}(x, b)$. Local efficiencies are substantially larger than the occupancy-weighted whole-cell values; the strongest local cells approach 10^{-1} while whole-cell values are typically 10^{-3}–10^{-2}.

5.5 State occupancy and support

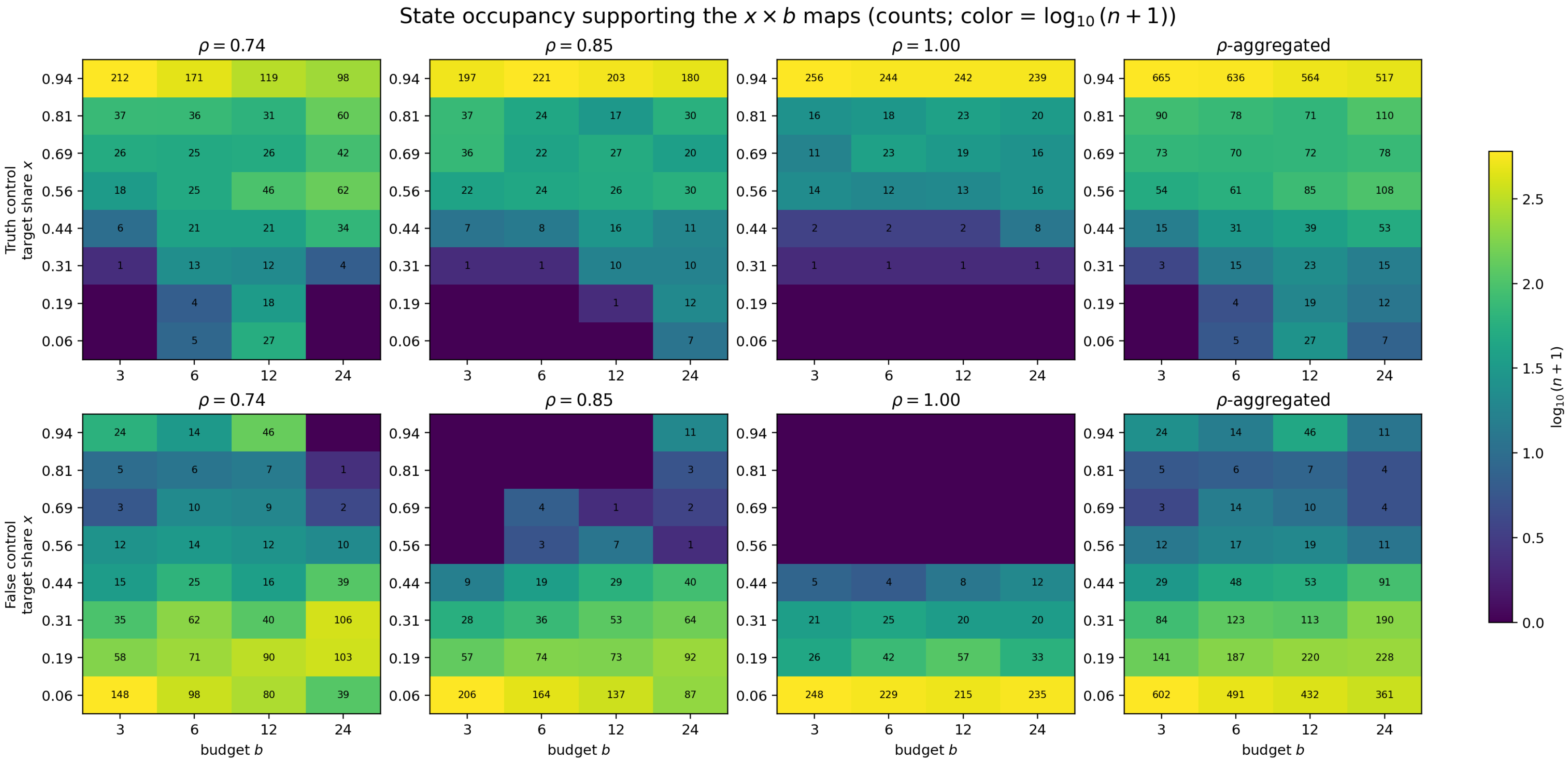

Figure 7: Observation counts underlying the state-resolved maps. The color is $\log_{10}(n+1)$ and the annotations give raw counts. Sparse states should not be interpreted with the same confidence as heavily occupied states.

6 Budget dependence and ρ-aggregated views

Figure 8 gives the same core metrics as lines in b, preserving the individual persistence slices and adding a simple equal-weight average across the three ρ values. This is useful for seeing non-monotonicity without losing the persistence-resolved information.

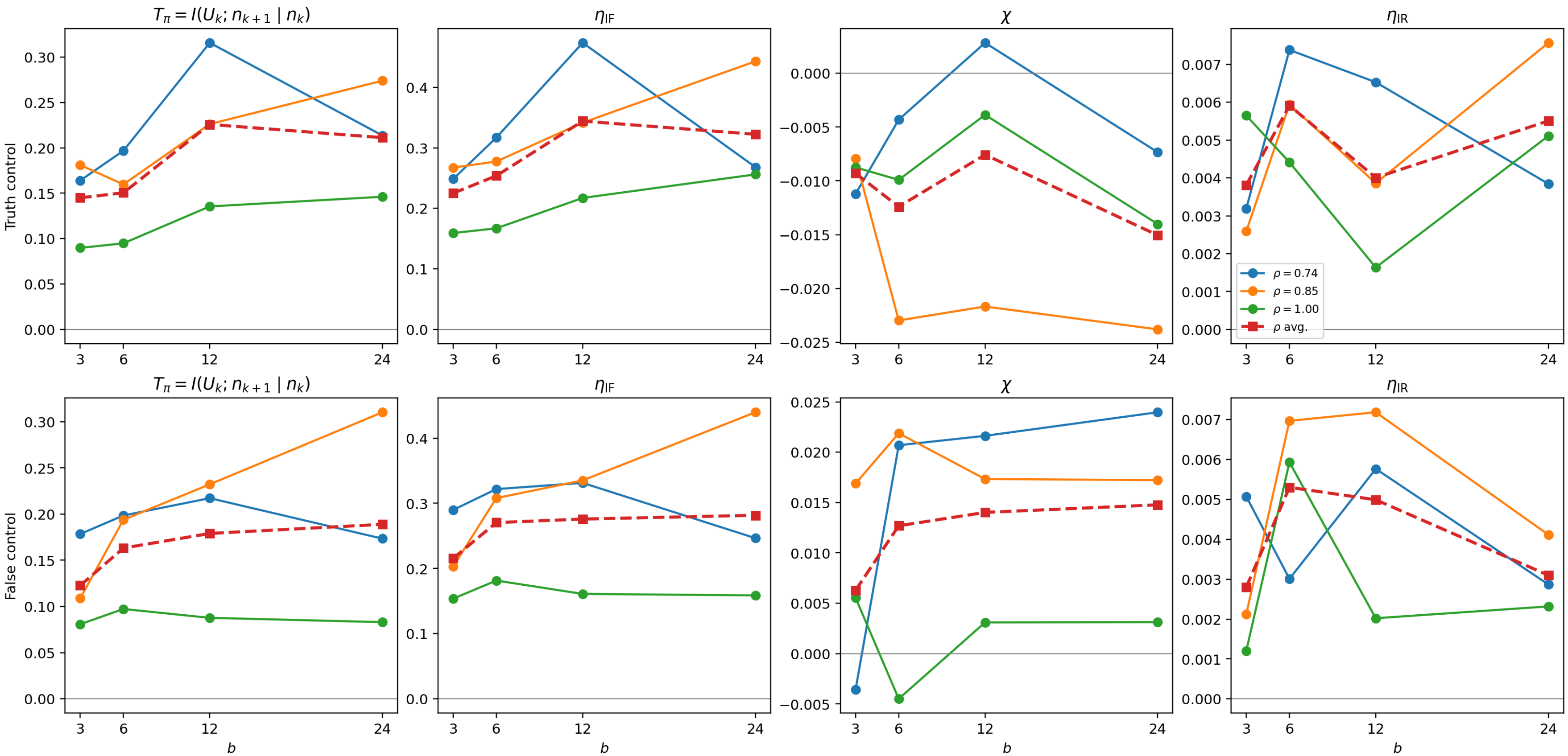

Figure 8: Whole-cell b dependence of T_π, η_{IF}, χ, and η_{IR}. Solid lines are individual persistence values; dashed square-marker lines are descriptive averages across ρ.

Figure 9 shows the final outcome more directly. At $\rho = 0.74$, the system is steerable but not monotonically so: false control is strongest at $b = 12$. At $\rho = 0.85$, high budget is required to produce substantial disruption. At $\rho = 1$, truth is effectively absorbing at the behavioral scale of this experiment.

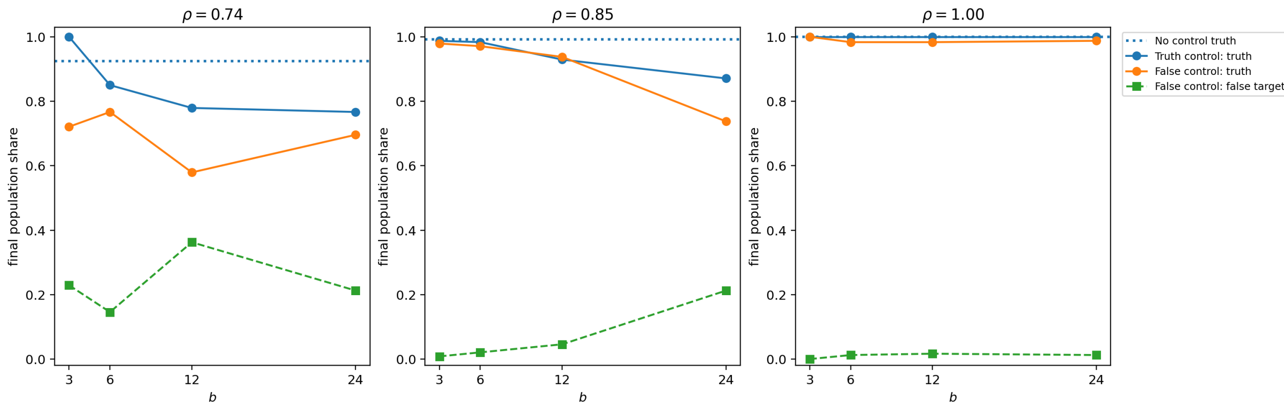

Figure 9: Final population shares versus b, separately for each persistence value. The dotted horizontal line is the no-control truth baseline at the same ρ.

7 Blackboard mechanism diagnostics

The blackboard actuator introduces a distinction between nominal budget and realized exposure. The coordinator seeds b DIRECTIVEs at dawn only when $U = 1$, but an ordinary focal agent is influenced by a DIRECTIVE only when it is sampled from the live board. Figure 10 summarizes that mechanism.

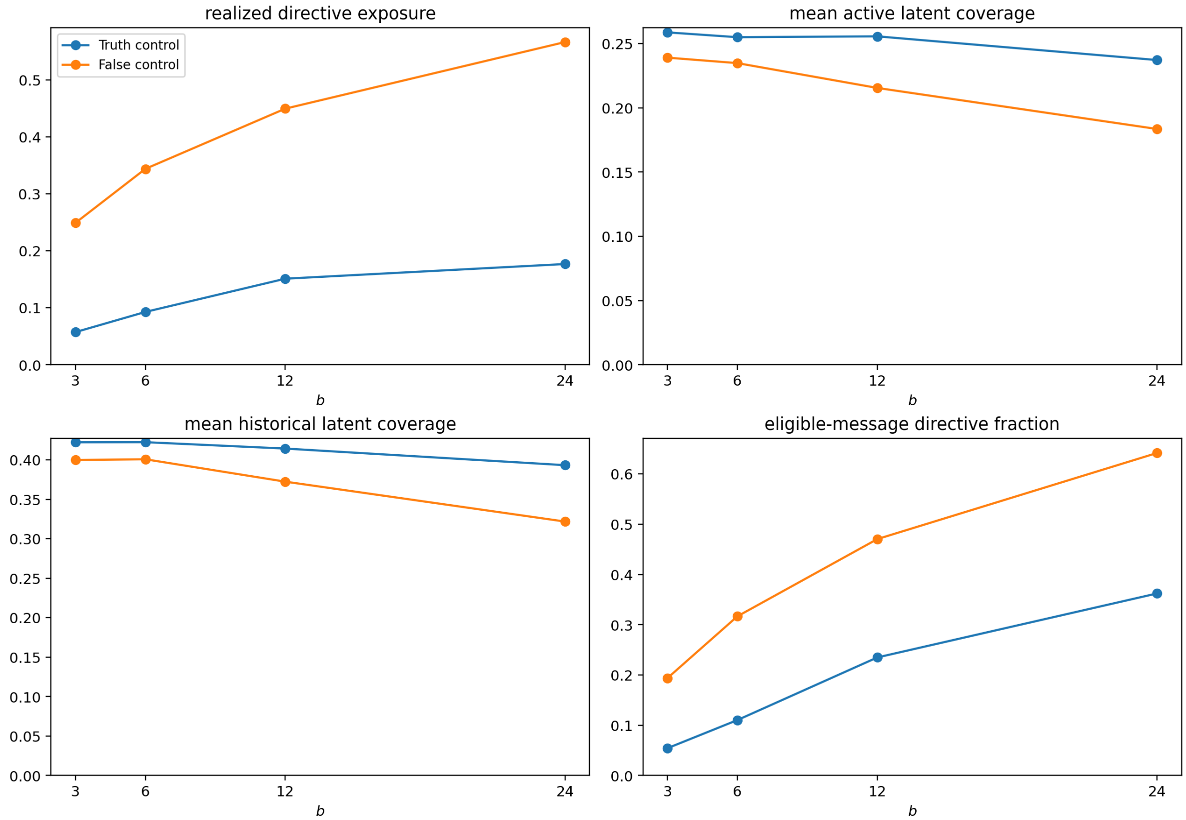

Figure 10: Blackboard diagnostics averaged over ρ for the two controlled arms. Larger b raises realized directive exposure, but is also associated with lower active/historical evidence coverage, consistent with competition for limited board attention.

This competition is particularly clear in the low-persistence truth-control cells. At $\rho = 0.74$, increasing b from 3 to 24 reduces mean active latent coverage from approximately 0.128 to 0.090 and reduces REPORT production while REQUEST production rises. Thus more controller presence can lower the throughput of evidence-bearing peer communication, providing a concrete mechanism for non-monotone control performance.

8 Temporal behavior at low persistence

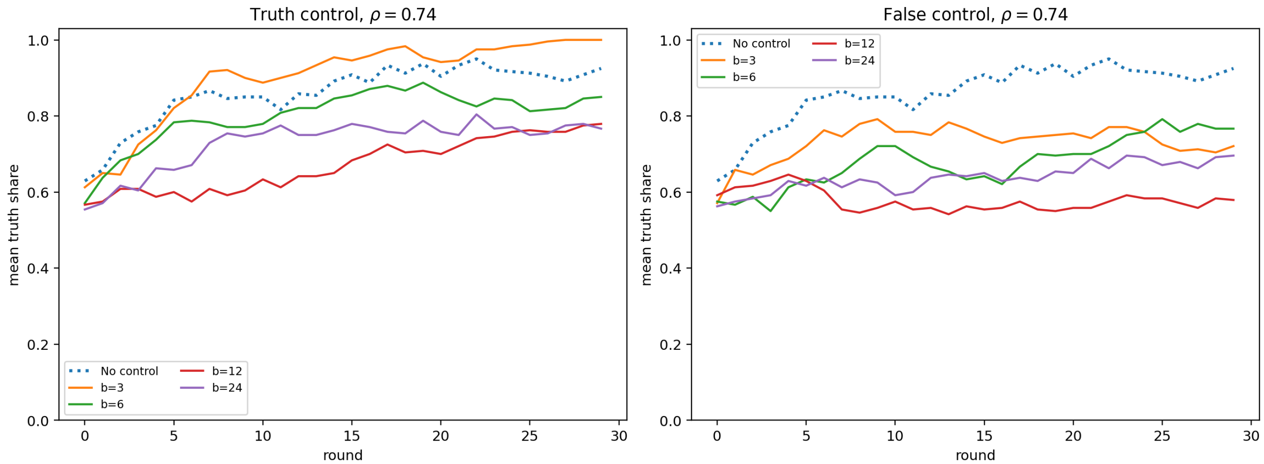

Figure 11: Mean truth-share trajectories at $\rho = 0.74$. Left: truth control. Right: false control. The no-control trajectory is shown as a common reference.

The trajectories support the interpretation that the low-persistence regime is the main current controllability window. The false controller can substantially delay or divert truth propagation, but only a minority of episodes end in actual false-target dominance. Truth-directed intervention can also over-control the system: the best final truth at $\rho = 0.74$ is obtained at the smallest tested budget, while larger budgets reduce performance.

9 Persistence crossover and the role of blackboard memory

The observed crossover is shifted toward lower persistence compared with the earlier peer-only persistence experiments. A natural mechanistic explanation is that nominal private-memory persistence ρ no longer controls all information retention. The blackboard creates a re-exposure channel:

$$K_i^{\text{active}} \xrightarrow{\text{deactivation}} K_i^{\text{hist}} \setminus K_i^{\text{active}} \xrightarrow{\text{REPORT re-exposure}} K_i^{\text{active}}.$$

Thus public communication can partially compensate for active forgetting. Operationally, the system behaves as if its usable evidence has a higher effective persistence than the bare ρ. The current grid only brackets this crossover; it does not locate a critical ρ precisely.

10 Thermodynamic status

The report intentionally preserves J_c because it is part of the established control-efficiency analysis. However, the current archive marks the affinity calibration as unsupported in every controlled cell. Consequently,

$$\eta_{\text{th}}, \qquad \eta_{\text{th}}^{\text{signed}}, \qquad hJ_c$$

are numerically undefined in this analysis package. The estimator records `insufficient_support_for_h_calibration`.

This is an analysis/theory issue introduced by the change from direct microscopic advocacy to the dawn-blackboard actuator, not evidence that the population experiment failed. The raw current J_c, sensing term, transfer information, susceptibility, and information-response efficiencies remain available. The next theoretical task is to determine the appropriate affinity/work calibration for the blackboard actuation channel rather than rerun the LLM experiment solely to obtain η_{th}.

11 Interpretation

1. **Truth is a strong endogenous attractor for task_001.** This explains the high no-control success and why global false-control capture is rare.
2. **Control nevertheless leaves a clear one-round signal.** T_π, η_{IF}, χ, and J_c vary systematically with x, b, and ρ.
3. **The response is state-local and non-monotone.** The $x \times b$ maps are more informative than global final accuracy or a single averaged efficiency.
4. **Lower persistence opens a controllable regime.** At $\rho = 0.74$, false steering can become substantial; at $\rho = 1$ it is essentially suppressed.
5. **The blackboard changes the phase structure.** Re-exposure and refresh provide additional effective memory, plausibly shifting the persistence crossover downward.
6. **Control can consume communication bandwidth.** At large b, DIRECTIVEs occupy more of the finite social channel and can reduce evidence-bearing peer communication, giving a natural mechanism for reduced efficiency at high control budgets.

12 Immediate next steps

The present experiment is already rich enough that the next priority should be analysis rather than expanding the grid:

- adapt the single-affinity h calibration for the dawn-blackboard actuator so that η_{th} can be defined without changing the empirical protocol;
- retain the $x \times b$ maps as the main efficiency view and use occupancy/support maps alongside them;
- if another persistence sweep is needed, concentrate extra ρ points below 0.85 to resolve the shifted controllability crossover;
- repeat the main grid on additional behaviorally validated MuSR tasks only after the task bank is frozen.

A Exploratory $q = 3$ false-control extension

The analysis archive includes 120 additional false-control episodes with $q = 3$. This extension is not part of the original 27-cell $q = 1$ design and is therefore treated as exploratory. It makes false steering markedly weaker despite greater controller-message exposure: only 1/120 episodes ends with the false target dominant. The likely mechanism is that larger social bandwidth also increases access to evidence-bearing peer messages, strengthening evidence recovery.

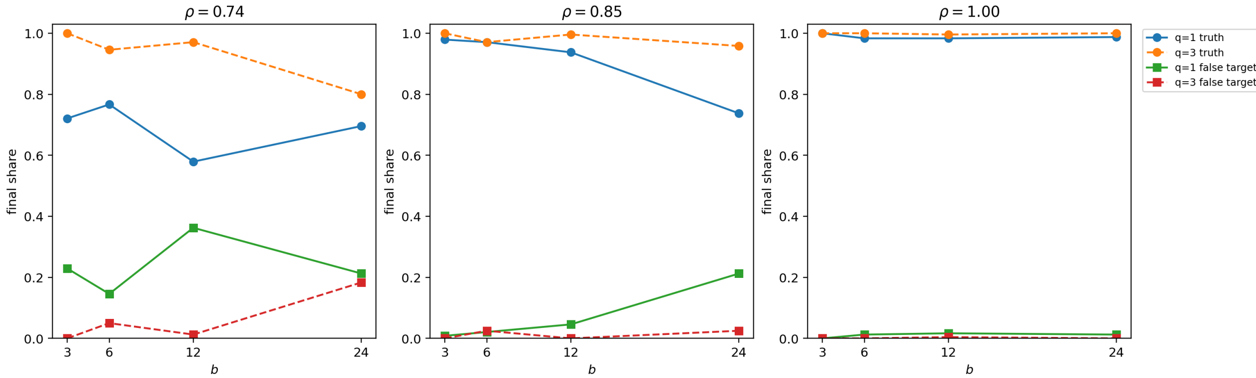

Figure 12: Exploratory comparison of false control with $q = 1$ and $q = 3$. Increasing social bandwidth strongly protects the truth outcome in this task.

B Whole-cell numeric reference

Table 4: Whole-cell control metrics for the 24 standard controlled cells ($q = 1$).

Arm	ρ	b	T_π	η_{IF}	χ	η_{IR}	J_c
Truth	0.74	3	0.164	0.249	-0.011	0.003	-2.78
Truth	0.74	6	0.197	0.317	-0.004	0.007	-0.72
Truth	0.74	12	0.316	0.474	0.003	0.007	1.65
Truth	0.74	24	0.214	0.268	-0.007	0.004	-1.12
Truth	0.85	3	0.181	0.267	-0.008	0.003	-1.98
Truth	0.85	6	0.160	0.277	-0.023	0.006	-5.70
Truth	0.85	12	0.226	0.342	-0.022	0.004	-4.12
Truth	0.85	24	0.274	0.443	-0.024	0.008	-6.20
Truth	1.00	3	0.090	0.159	-0.009	0.006	-2.50
Truth	1.00	6	0.095	0.167	-0.010	0.004	-1.94
Truth	1.00	12	0.135	0.217	-0.004	0.002	-0.83
Truth	1.00	24	0.146	0.256	-0.014	0.005	-3.52
False	0.74	3	0.178	0.290	-0.004	0.005	-2.06
False	0.74	6	0.198	0.322	0.021	0.003	11.07
False	0.74	12	0.217	0.331	0.022	0.006	13.06
False	0.74	24	0.173	0.247	0.024	0.003	12.37
False	0.85	3	0.109	0.203	0.017	0.002	9.98
False	0.85	6	0.194	0.308	0.022	0.007	11.19
False	0.85	12	0.232	0.335	0.017	0.007	8.76
False	0.85	24	0.310	0.440	0.017	0.004	8.62
False	1.00	3	0.081	0.154	0.006	0.001	3.36
False	1.00	6	0.097	0.181	-0.004	0.006	-3.16
False	1.00	12	0.088	0.161	0.003	0.002	1.73
False	1.00	24	0.083	0.159	0.003	0.002	1.37

C Estimator and support notes

The source analysis reports 11,700 round records and 280,800 micro-slot records across all 390 episodes, with no failed or aborted episodes. Information quantities use the repository's established direct-counting round-feedback estimator, whole-episode bootstrap, configured nulls, and support diagnostics. State-local cells marked unsupported are omitted from the reconstructed heatmaps. The ρ-aggregated $x \times b$ panels are observation-weighted descriptive summaries, not new pooled inferential estimators.

The validation archive reports no structural errors and warns only that 12,160 superseded trajectory records from safe retries were excluded.